MYP MATHEMATICS 1 · SECOND EDITION

03

Factors, multiples and primes

Part I · The grammar of number

CHAPTER 3 · OPENER

Factors, multiples and primes

Factors, multiples and prime products reveal how numbers are built, so exact grouping and shared schedules can be planned and justified.

ENGAGEMENT

Can 192 students be placed into identical mixed teams without leaving anyone out? What must be true about the number of teams?

Learning goals

- List factors and multiples systematically.
- Apply divisibility tests and classify primes.
- Write prime products and find HCF and LCM.
- Choose a grouping or schedule method and justify it.

MYP lens

Key concept: Relationships

Related concepts: System, Representation

Global context: Fairness and development

ATL focus. In order for a student to justify a generalisation, students must organise evidence, test a proposed rule and explain why it works beyond the examples.

Language bank: factor · multiple · factor pair · divisible · prime · composite · prime factorisation · HCF · LCM

CHAPTER 3 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the chapter map from concrete examples to explanations and decisions.

READY CHECK

Try these without a calculator. They check the ideas this chapter builds on.

R1	Calculate 7×8 and $96 \div 8$.	READY
R2	List the first five positive multiples of 4.	READY
R3	Write 100 as a product of two equal factors.	READY
R4	Which is larger, 3^2 or 2^3?	READY
R5	Does 7 divide 56 exactly? Explain.	READY

Chapter map

1. **3.1 Factors, multiples and factor pairs** — Search for factor pairs in order, then stop when the partners would repeat.
2. **3.2 Divisibility tests** — Use digit tests as shortcuts, then confirm with an exact multiplication when needed.
3. **3.3 Prime and composite numbers** — Classify by the number of factors, and stop a prime search at the square-root boundary.
4. **3.4 Prime factorisation** — Break a composite into factors until every branch ends at a prime.
5. **3.5 Highest common factor and lowest common multiple** — Choose HCF for largest equal groups and LCM for the next shared repeat.

> **EXPLAIN HOW YOU KNOW**
>
> For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 3 · MATHEMATICAL DEVELOPMENT

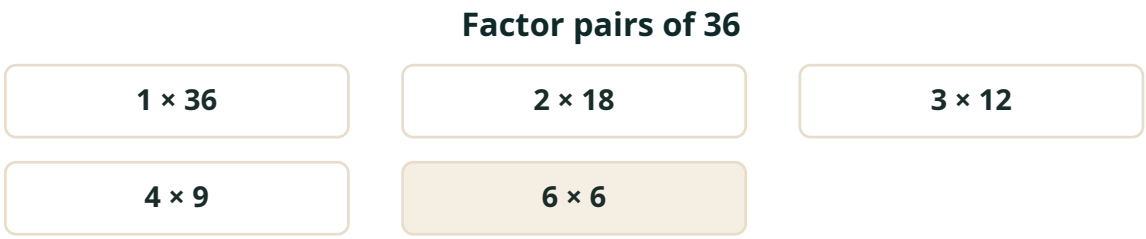
3.1 Factors, multiples and factor pairs

BIG IDEA

Search for factor pairs in order, then stop when the partners would repeat.

A factor divides a positive whole number exactly. A multiple is that number times a whole number. Since $7 \times 12 = 84$, 7 and 12 are factors of 84, and 84 is a multiple of both.

For 36, the pairs are 1×36, 2×18, 3×12, 4×9 and 6×6. The repeated 6 is counted once. A pair search is complete when the next possible first factor has already appeared as a partner.



Factor pairs meet at 6×6. Search in order so no pair is missed.

WORKED EXAMPLE · FIND EVERY FACTOR OF 48

Use factor pairs rather than guessing.

1. 1×48, 2×24, 3×16, 4×12 and 6×8.
2. 5 and 7 do not divide 48. The next partner would repeat.
3. Collect the factors in order.

1, 2, 3, 4, 6, 8, 12, 16, 24, 48: ten factors.

CHAPTER 3 · PRACTICE

3.1 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

A factor is usually no larger than the number; a positive multiple is at least as large. Do not confuse the two directions.

GUIDED PRACTICE · FIRST STEPS

3.1.1 List all factor pairs of 36. FLUENCY

3.1.2 List the first five positive multiples of 7. FLUENCY

3.1.3 Is 56 a multiple of 8? Is 8 a factor of 56? Explain. FLUENCY

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

3.1.4 A student lists 1, 2, 3, 6, 9, 18 as all factors of 54. Find the missing factors. REASONING

3.1.5 Which rectangular arrays can be made with exactly 24 counters? Count rotations as the same array. APPLICATION

3.1.6 A number between 30 and 50 has factors 6 and 8. Find it and explain uniqueness. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 3 · MATHEMATICAL DEVELOPMENT

3.2 Divisibility tests

> **BIG IDEA**
> Use digit tests as shortcuts, then confirm with an exact multiplication when needed.

Tests for 2, 5 and 10 use the last digit. Tests for 4 and 8 use the last two and three digits because 100 is divisible by 4 and 1,000 by 8.

For 3 and 9, add the digits. A number is divisible by 6 exactly when it is divisible by both 2 and 3. These tests concern positive whole numbers here; a passing test is evidence, not a guess.

WORKED EXAMPLE · TEST 4,728

Which of 2, 3, 4, 5, 6, 8, 9 and 10 divide it?

1. Last digit 8: passes 2, fails 5 and 10. Digit sum 21: passes 3, fails 9.
2. Last two digits 28: passes 4. Last three digits 728 = 8 × 91: passes 8.
3. Passing 2 and 3 means it also passes 6.

2, 3, 4, 6 and 8.

CHAPTER 3 · PRACTICE

3.2 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

A digit sum divisible by 3 does not prove divisibility by 6: the number must also be even.

GUIDED PRACTICE · FIRST STEPS

3.2.1 Which of 2, 3, 4, 5, 6, 8, 9 and 10 divide 4,728? FLUENCY

3.2.2 Test 315 for divisibility by 3, 5, 9 and 10. FLUENCY

3.2.3 Explain why 513 is not divisible by 6 even though its digit sum is 9. REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

3.2.4 Why does testing the last two digits decide divisibility by 4? REASONING

3.2.5 A teacher has 126 pencils. Can they be shared equally into 6 packs? Into 9 packs? APPLICATION

3.2.6 Use digits 4, 5 and 6 once each to make the largest three-digit number divisible by 4. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 3 · MATHEMATICAL DEVELOPMENT

3.3 Prime and composite numbers

BIG IDEA

Classify by the number of factors, and stop a prime search at the square-root boundary.

A prime is a whole number greater than 1 with exactly two positive factors: 1 and itself. A composite has more than two. The number 1 has one factor, so it is neither.

To test n, try prime divisors no larger than √n. If n = a × b and both factors exceeded √n, their product would exceed n. Thus a composite must have a factor at or below that boundary.

WORKED EXAMPLE · IS 97 PRIME?

Test only the primes up to √97.

1. √97 is between 9 and 10, so test 2, 3, 5 and 7.
2. 97 is not divisible by any of them.

97 is prime.

CHAPTER 3 · PRACTICE

3.3 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

2 is prime even though it is even. It is the only even prime; every larger even number has 2 as a factor.

GUIDED PRACTICE · FIRST STEPS

3.3.1 List the primes below 30. FLUENCY

3.3.2 Classify 1, 2, 21 and 29 as prime, composite or neither. FLUENCY

3.3.3 Which prime divisors must be tested to decide whether 97 is prime? Is it prime? REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

3.3.4 Give a counterexample to “all odd numbers are prime”. REASONING

3.3.5 Find a twin-prime pair between 50 and 70 and give its sum. APPLICATION

3.3.6 Does checking that 28 = 11 + 17 prove that every even number above 2 is a sum of two primes? CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 3 · MATHEMATICAL DEVELOPMENT

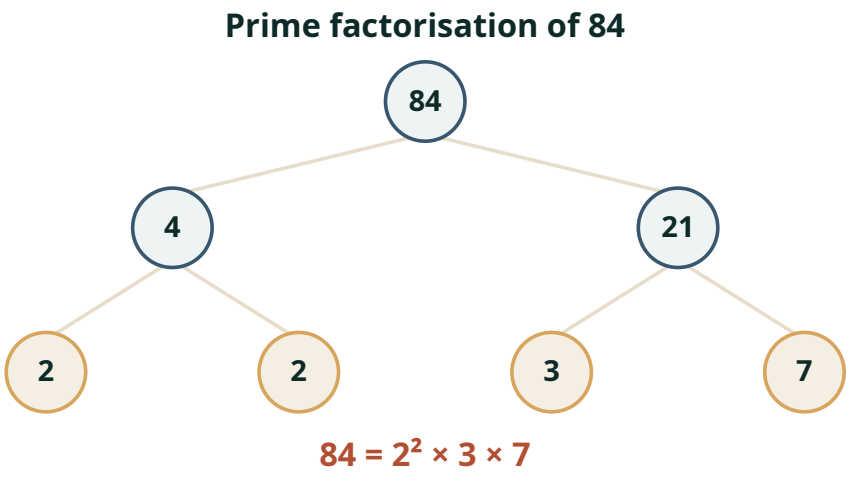
3.4 Prime factorisation

BIG IDEA

Break a composite into factors until every branch ends at a prime.

A factor tree records repeated splitting. Different starting pairs can lead to the same prime factors, apart from order. Index form groups equal primes.

For 84, split 84 into 4×21, then 4 into 2×2 and 21 into 3×7. Thus $84 = 2^2 \times 3 \times 7$. Multiply the prime product to check the original number.



Split composite branches until every endpoint is prime: $84 = 2^2 \times 3 \times 7$.

WORKED EXAMPLE · FACTORISE 72

Use a factor tree or repeated division.

1. $72 = 8 \times 9$.
2. $8 = 2^3$ and $9 = 3^2$.
3. Check $8 \times 9 = 72$.

$72 = 2^3 \times 3^2$.

CHAPTER 3 · PRACTICE

3.4 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

Stop only when all terminal factors are prime. A product containing 4 or 9 is not yet a prime factorisation.

GUIDED PRACTICE · FIRST STEPS

3.4.1 Write 36, 54 and 72 as products of primes in index form. FLUENCY

3.4.2 Find the number represented by $2^3 \times 3 \times 5$. FLUENCY

3.4.3 A tree for 84 ends in 4, 3 and 7. Is it complete? Repair it. REASONING

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

3.4.4 Why can two different factor trees for 84 give the same final prime product? REASONING

3.4.5 A display has 126 tiles. Write 126 as a prime product and use it to find a 6-by-21 array. APPLICATION

3.4.6 Find the smallest positive integer that makes 180n a perfect square. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 3 · MATHEMATICAL DEVELOPMENT

3.5 Highest common factor and lowest common multiple

BIG IDEA

Choose HCF for largest equal groups and LCM for the next shared repeat.

The highest common factor (HCF) is the largest number dividing each given number. The lowest common multiple (LCM) is the smallest positive number that is a multiple of each.

Using prime products, take the smallest exponent of each shared prime for the HCF and the largest exponent of each prime appearing for the LCM. For $12 = 2^2 \times 3$ and $18 = 2 \times 3^2$, HCF = 6 and LCM = 36.

WORKED EXAMPLE · RIBBONS AND SCHEDULES

48 cm and 60 cm ribbons are cut into the longest equal pieces. Two bells repeat every 15 and 20 minutes.

1. HCF(48,60) = 12, so pieces are 12 cm long.
2. There are 4 + 5 = 9 pieces.
3. LCM(15,20) = 60, so the bells next coincide after 60 minutes.

12 cm, 9 pieces; 60 minutes.

CHAPTER 3 · PRACTICE

3.5 Guided and independent practice

WATCH FOR THIS MISCONCEPTION

HCF cannot exceed the smaller number; a positive LCM cannot be below the larger number. Use these bounds to check.

GUIDED PRACTICE · FIRST STEPS

3.5.1 Find the HCF and LCM of 12 and 18. FLUENCY

3.5.2 Find the HCF and LCM of 8, 12 and 18. FLUENCY

3.5.3 Cut 48 cm and 60 cm ribbons into the longest equal pieces with no waste. Find the length and total pieces. APPLICATION

INDEPENDENT PRACTICE · EXPLAIN AND APPLY

3.5.4 Buses leave every 15 and 20 minutes and coincide at 08:00. When do they next coincide? APPLICATION

3.5.5 A student uses HCF to find when two repeating events coincide. Explain the correct choice. REASONING

3.5.6 Find two different unordered pairs of positive integers with HCF 6 and LCM 36. CHALLENGE

Check one answer with a different representation or an inverse operation. Record the check in your working.

CHAPTER 3 · EXPLORATION

Why do squares have an odd factor count?

Count factors of positive whole numbers. A repeated factor such as 6 in 6 × 6 is counted once.

> **INVESTIGATION**
> Criterion B. Record evidence, describe a pattern, then justify it using factor pairs.

N	FACTOR PAIRS	NUMBER OF FACTORS
4	1×4; 2×2	3
9		
16		
25		
36		

EXPLORATION · WHY DO SQUARES HAVE AN ODD FACTOR COUNT?

Criterion B. Record evidence, describe a pattern, then justify it using factor pairs.

B1 Complete the factor counts for 4, 9, 16, 25 and 36. INVESTIGATION

B2 Which numbers from 1 to 30 have an odd number of factors? INVESTIGATION

B3 Explain why a non-square has an even number of factors. INVESTIGATION

B4 Verify the square pattern on 49 and compare factor counts of 24 and 25. INVESTIGATION

B5 Find every number below 100 with exactly three factors. Explain the rule. CHALLENGE

CHAPTER 3 · CHECKPOINT

Checkpoint and reflection

CHECKPOINT

Answer, then explain one result to a partner.

C1	List all factors of 24.	CHECK
C2	Is 91 prime?	CHECK
C3	Write 90 as a prime product.	CHECK
C4	Find HCF(18,24) and LCM(18,24).	CHECK

Reflection

- Which representation made the idea clearest: a diagram, a table, symbols or words?
- Where did you check an answer instead of trusting it?
- Write one question you would ask a partner to test understanding.

CHAPTER 3 · APPLICATION

The fairness relay

A simulated school festival has 84, 60 and 48 students in three year groups. Every student takes part exactly once. Identical mixed teams must contain the same number from each year group.

> **REAL-LIFE APPLICATION**
> Criterion D. Simulated festival data. State what fairness means before choosing a grouping.

YEAR GROUP	STUDENTS
A	84
B	60
C	48

APPLICATION · THE FAIRNESS RELAY

Criterion D. Simulated festival data. State what fairness means before choosing a grouping.

D1 What is the largest possible number of identical mixed teams? How many students are in each? APPLICATION

D2 If instead each year forms teams of 12, how many teams are there altogether? APPLICATION

D3 Both plans include everyone. Explain why “fair” needs a definition. REASONING

D4 Could 16 identical mixed teams be made? Explain. CHALLENGE

CHAPTER 3 · ASSESSMENT PRACTICE

The number museum

MYP ASSESSMENT PRACTICE

Criteria A and C. Create a visitor card for 84 and a second number, 60. Use precise language, a factor tree and a short explanation.

Your display should help a younger student see the structure of a number, not merely list calculations.

ASSESSMENT PRACTICE · THE NUMBER MUSEUM

Criteria A and C. Create a visitor card for 84 and a second number, 60. Use precise language, a factor tree and a short explanation.

S1 Show all factor pairs of 84 and write its prime factorisation. ASSESSMENT

S2 Find HCF(84,60) and LCM(84,60), and explain which answers “largest equal groups?”. ASSESSMENT

S3 Write a short museum label and one visitor question with its answer. ASSESSMENT

Success criteria

- Use correct mathematical language and notation.
- Show a method or representation that another student can follow.
- Check the result and explain why it is reasonable.

CHAPTER 3 · MIXED PRACTICE

Mixed practice 3

Connect ideas across the chapter. Give units and reasons where requested.

MIXED PRACTICE 3

Connect ideas. Give units and reasons where requested.

M1 List the factors of 42. FLUENCY

M2 Write the first six positive multiples of 9. FLUENCY

M3 Which of 2,3,4,5,6,8,9,10 divide 360? FLUENCY

M4 Explain why 1 is neither prime nor composite. REASONING

M5 Prime-factorise 96 and 150. FLUENCY

M6 Two lights flash every 18 and 24 seconds. After how many seconds do they next flash together? APPLICATION

M7 Find the longest equal pieces that can be cut from 42 cm and 70 cm strings without waste. APPLICATION

M8 Does a larger number always have more factors? Compare 28 and 29. REASONING

M9 Find the smallest number with prime factorisation containing 2^2, 3 and 5, then list its factor count. CHALLENGE

M10 Two numbers have HCF 4 and LCM 60. One number is 12. Find the other and check. CHALLENGE

FINAL REFLECTION

Choose one item you solved confidently and one that needed a revision. Explain the difference in your method.

CHAPTER 3 · ANSWERS AND REASONING

Answers · Ready check and Checkpoint

READY CHECK

R1 **56 and 12.**
Use the inverse multiplications 8×7 and 8×12.

R2 **4, 8, 12, 16, 20.**
Multiply by 1 through 5.

R3 **10×10.**
10 is the square root of 100.

R4 **$3^2 = 9$ is larger than $2^3 = 8$.**
Evaluate each power.

R5 **Yes: $7 \times 8 = 56$.**
The remainder is zero.

CHECKPOINT

C1 **1, 2, 3, 4, 6, 8, 12, 24.**
Use four factor pairs.

C2 **No: $91 = 7 \times 13$.**
Test primes through $\sqrt{91}$, including 7.

C3 **$2 \times 3^2 \times 5$.**
$90 = 9 \times 10$.

C4 **HCF 6; LCM 72.**
$18 = 2 \times 3^2$; $24 = 2^3 \times 3$.

CHAPTER 3 · ANSWERS AND REASONING

Answers · 3.1 and 3.2

PRACTICE 3.1 · FACTOR PAIRS

3.1.1 **1 × 36; 2 × 18; 3 × 12; 4 × 9; 6 × 6.**
Each product is 36; the final pair repeats a factor.

3.1.2 **7, 14, 21, 28, 35.**
Multiply 7 by 1, 2, 3, 4 and 5.

3.1.3 **Yes to both: 8 × 7 = 56.**
The same exact multiplication links factors and multiples.

3.1.4 **Missing: 27 and 54. The pairs include 2 × 27 and 1 × 54.**
Search pairs systematically; do not stop at the square root without recording both partners.

3.1.5 **1 × 24, 2 × 12, 3 × 8 and 4 × 6.**
Each pair of side lengths has product 24.

3.1.6 **48. It is the only common multiple of 6 and 8 in the interval.**
List common multiples or use the least common multiple 24, then 48.

PRACTICE 3.2 · DIVISIBILITY

3.2.1 **2, 3, 4, 6 and 8.**
Use the last digits and digit sum 21; 728 is divisible by 8.

3.2.2 **3, 5 and 9: yes. 10: no.**
Digit sum 9; last digit 5, not 0.

3.2.3 **It is odd, so it fails the test for 2. Divisibility by 6 needs both 2 and 3.**
The digit-sum test establishes only the factor 3 here.

3.2.4 **Every hundred is divisible by 4, so only the last two digits can affect the remainder.**
Write the number as 100 × a + b, where b is the last two-digit number.

3.2.5 **6 packs: yes, 21 per pack. 9 packs: yes, 14 per pack.**
126 is even and has digit sum 9, so both tests pass.

3.2.6 **654 is not divisible by 4; the largest valid arrangement is 564.**
The possible even endings are 54, 64 and 46; only 64 and 56 work, and 564 is the largest valid arrangement.

CHAPTER 3 · ANSWERS AND REASONING

Answers · 3.3 and 3.4

PRACTICE 3.3 · PRIMES

3.3.1 **2, 3, 5, 7, 11, 13, 17, 19, 23, 29.**
Each has exactly two positive factors.

3.3.2 **1 neither; 2 prime; 21 composite; 29 prime.**
21 = 3 × 7; 1 has only one factor.

3.3.3 **Test 2, 3, 5 and 7. None divides 97, so it is prime.**
The next prime, 11, is greater than √97.

3.3.4 **9 is odd but composite: 9 = 3 × 3.**
One counterexample disproves a universal claim.

3.3.5 **59 and 61; sum 120.**
Both are prime and differ by 2.

3.3.6 **No. It verifies one case, not a general proof.**
A general claim needs a reason covering every case.

PRACTICE 3.4 · PRIME PRODUCTS

3.4.1 **$36 = 2^2 \times 3^2$; $54 = 2 \times 3^3$; $72 = 2^3 \times 3^2$.**
Split until every factor is prime, then group equal factors.

3.4.2 **120.**
8 × 3 × 5 = 120.

3.4.3 **No: 4 is composite. Replace it by 2 × 2, giving $2^2 \times 3 \times 7$.**
A prime factorisation cannot contain 4 as a terminal factor.

3.4.4 **Prime factorisation is unique apart from order; the trees only choose different intermediate pairs.**
Both trees decompose the same number completely.

3.4.5 **$126 = 2 \times 3^2 \times 7 = 6 \times 21$.**
Group 2 × 3 as 6 and 3 × 7 as 21.

3.4.6 **n = 5, because $180 = 2^2 \times 3^2 \times 5$ and $900 = 30^2$.**
A square has even exponents in its prime factorisation.

CHAPTER 3 · ANSWERS AND REASONING

Answers · 3.5

PRACTICE 3.5 · HCF AND LCM

3.5.1 **HCF 6; LCM 36.**
Shared prime factors give 6; the smallest common multiple is 36.

3.5.2 **HCF 2; LCM 72.**
2 divides all three; 72 is the first positive multiple shared by all.

3.5.3 **12 cm; 9 pieces in total.**
HCF is 12; 48÷12 + 60÷12 = 4 + 5.

3.5.4 **09:00, after 60 minutes.**
The next shared repeat is LCM(15,20) = 60 minutes.

3.5.5 **Use LCM: a shared future time must be a common multiple, and the next one is the lowest.**
HCF describes a largest common divisor, not a future repeat.

3.5.6 **(6,36) and (12,18).**
Write the numbers as 6a and 6b with coprime a,b and ab = 6.

CHAPTER 3 · ANSWERS AND REASONING

Answers · Exploration

EXPLORATION · WHY DO SQUARES HAVE AN ODD FACTOR COUNT?

B1 **3, 3, 5, 3, 9.**
List pairs and count a repeated middle factor once.

B2 **1, 4, 9, 16 and 25: the perfect squares.**
Each square has one pair whose two entries coincide.

B3 **Its factors form distinct pairs a × b, with a different from b.**
No unpaired middle factor occurs.

B4 **49 has 3 factors. 24 has 8; 25 has 3. A larger number need not have more factors.**
$49 = 7^2$; compare complete factor lists.

B5 **4, 9, 25 and 49: squares of primes. p^2 has factors 1, p and p^2.**
A three-factor number must have one repeated middle pair and no other prime factor.

CHAPTER 3 · ANSWERS AND REASONING

Answers · Application and Assessment

APPLICATION · THE FAIRNESS RELAY

D1 **12 teams of 16: 7 from A, 5 from B and 4 from C.**
The team count must divide all three totals; use HCF.

D2 **7 + 5 + 4 = 16 teams.**
Divide each year total by 12.

D3 **Equal mixed composition and equal team size within each year are different fairness rules. Both can be valid for different purposes.**
State the criterion before judging the allocation.

D4 **No. 16 does not divide 84, 60 or 48 exactly.**
A feasible team count must be a common factor.

ASSESSMENT PRACTICE · THE NUMBER MUSEUM

S1 **1×84, 2×42, 3×28, 4×21, 6×14, 7×12; $84 = 2^2 \times 3 \times 7$.**
There are twelve factors; the prime product checks to 84.

S2 **HCF 12; LCM 420. HCF answers largest equal groups.**
Use shared prime factors for HCF and highest powers for LCM.

S3 **Sample: “84 can be arranged in six different rectangular arrays. Its prime building blocks are 2, 2, 3 and 7.” Question: “How many factors does 84 have?” Answer: 12.**
A clear label connects a representation to a verified fact.

CHAPTER 3 · ANSWERS AND REASONING

Answers · Mixed practice

MIXED PRACTICE 3

M1 **1, 2, 3, 6, 7, 14, 21, 42.**
Pair 1×42, 2×21, 3×14 and 6×7.

M2 **9, 18, 27, 36, 45, 54.**
Multiply 9 by 1 through 6.

M3 **All eight listed numbers.**
360 is a common multiple of each.

M4 **It has exactly one positive factor, not two or more than two.**
Prime and composite definitions both apply to numbers greater than 1.

M5 **$96 = 2^5 \times 3$; $150 = 2 \times 3 \times 5^2$.**
Group repeated primes.

M6 **72 seconds.**
Use LCM, not HCF.

M7 **14 cm.**
Use HCF(42,70).

M8 **No. 28 has 6 factors; 29 is prime and has 2.**
A prime can be larger than a highly divisible composite.

M9 **60; it has 12 factors.**
Multiply 4×3×5; exponents 2,1,1 give (2+1)(1+1)(1+1).

M10 **20. HCF(12,20)=4 and LCM(12,20)=60.**
For two positive integers, HCF × LCM equals their product.